

DynLLM: Integrating LLM-Based Profiles with Dynamic Graph Recommendation (Example-Rich Version)

DynLLM (Zhao et al., 2024) proposes a recommendation model that combines language-model-derived semantic profiles with temporal dynamic graph embeddings to make next-item predictions. It is specifically designed for continuous-time dynamic graphs (CTDGs) such as user-item interaction networks in e-commerce platforms.

LLM-Based User Profiling (With Examples)

DynLLM uses a large language model (LLM) to summarize a user's preferences based on the titles of previously interacted items.

Example: Suppose a user's recent interactions include:

1. "Canon EOS M200 Mirrorless Camera"
2. "Beginner's Photography Guidebook"
3. "Nike Air Zoom Pegasus 39 Running Shoes"

The system feeds these titles to an LLM using a prompt like:

"Based on the user's purchase history, identify their crowd segment, personal interest, preferred product categories, and preferred brands."

LLM Output (Example):

- Crowd segment: Tech-savvy hobbyist
- Personal interest: Photography, Fitness
- Categories: Electronics, Sportswear
- Brands: Canon, Nike

Each of these four facets is converted into a vector embedding using a sentence embedding model (like `text-embedding-ada-002`), yielding four separate semantic embeddings per user:

p_1 (crowd), p_2 (interest), p_3 (category), p_4 (brand), each of size 128.

Temporal Graph Embedding with TGAN (With Examples)

The user's interactions are modeled as edges in a time-evolving bipartite graph, where nodes are users and items, and edges represent interactions with timestamps.

Example:

- T1: U123 $\rightarrow$ Canon EOS M200
- T2: U123 $\rightarrow$ Nike Shoes
- T3: U123 $\rightarrow$ Fitbit Charge 5

This results in the following edges:

U123 --(T1)--> Camera

U123 --(T2)--> Running Shoes

U123 --(T3)--> Fitness Tracker

These interactions are fed to a Temporal Graph Attention Network (TGAN), which:

- Uses attention to weigh recent interactions more heavily
- Learns a time-sensitive user embedding, u_{graph} , updated at each interaction

Example Embedding Evolution:

- After T1: $u_{\text{graph}} = [0.02, -0.03, \dots, 0.10]$
- After T2: $u_{\text{graph}} = [0.05, 0.00, \dots, 0.13]$
- After T3: $u_{\text{graph}} = [0.09, 0.01, \dots, 0.17]$

TGAN considers not only what the user interacted with but also when, giving more emphasis to recent behaviors (e.g., $T3 > T1$).

Multi-Faceted Distilled Attention Fusion (With Examples)

Now, the model has:

- u_{graph} : user's embedding from the dynamic graph
- p_1, p_2, p_3, p_4 : four profile embeddings from the LLM

To combine them, DynLLM uses a multi-faceted distilled attention mechanism.

Step 1: Learn attention weights

For each user, learn how important each profile facet is:

Example:

attention = [crowd: 0.10, interest: 0.40, category: 0.25, brand: 0.25]

Interpretation: This user's interests (e.g., photography, fitness) are more informative than general demographic or brand preference.

Step 2: Create a distilled LLM profile

Weighted sum of the four profile vectors:

$$p_{\text{final}} = 0.10 * p_1 + 0.40 * p_2 + 0.25 * p_3 + 0.25 * p_4$$

Resulting in a single 128-dim profile vector enriched with semantic preferences.

Step 3: Fuse with the graph embedding

Combine p_{final} and u_{graph} :

Example: Concatenate $\rightarrow [p_{\text{final}} || u_{\text{graph}}] \rightarrow 256\text{-dim vector}$

Then pass through a feedforward network (MLP) to obtain the final user embedding u_{final} .

Recommendation (With Example)

Given the fused representation u_{final} , the model scores all candidate items for the next interaction.

Example Output: Predicted top-3 items

1. "Canon EF 50mm f/1.8 STM Lens"
2. "Adidas Lightweight Training Shorts"
3. "Camera Tripod Stand"

Interpretation:

The system recommends products related to photography and fitness, balancing the user's interest vector with their recent purchases.

Evaluation and Results

DynLLM was tested on two real-world datasets from Tmall and Alibaba and achieved superior performance over static GNNs (GCN, LightGCN) and dynamic baselines (TGN, JODIE).

Ablation studies confirmed:

- Without the LLM profiles $\rightarrow$ performance drops

- Without the attention mechanism → fusion becomes noisy and less effective

Why It Matters for the Survey

DynLLM is among the first to explicitly integrate LLM-generated semantic signals into a dynamic graph learning pipeline. It addresses the incompleteness of structure-only models by injecting natural language-derived user intent. The distilled attention mechanism makes the LLM integration robust to noise and adaptable, making this paper highly relevant for research on combining LLMs with dynamic GNNs.